MolSanity: A Reliability Audit for Feature Attributions on Molecular Graph Neural Networks

Karthik Iyer[1,*] Nasser Sabar[1]

[1]Department of Computer Science and Information Technology, La Trobe University, Melbourne, Victoria 3086, Australia

*Correspondence: karthik.iyer@hotmail.com ORCID: 0009-0004-0593-1602

Abstract. Feature attributions are routinely used to justify molecular graph neural network (GNN) predictions to chemists, yet they are almost never audited for *reliability*: existing evaluation frameworks ask whether an explanation is faithful to the model, not whether the model's explanation can be trusted, and not where it stops being trustworthy under the scaffold shift that characterises real drug-discovery workflows. We present **MolSanity**, a reliability-audit framework that wraps canonical attribution implementations (Captum, PyTorch Geometric, RDKit) rather than proposing a new attributor, and scores every (dataset × backbone × attributor × split) cell on six axes: motif-native coherence, occlusion–attribution faithfulness, ground-truth localisation where node labels exist, cross-checkpoint stability, calibration linkage, and confidence/correctness regime stratification. Our central finding is that *faithfulness is not correctness*, and that the two invert under shift. Holding dataset, backbone and attributor set fixed and changing only the split, all 3 faithfulness metrics we test (our occlusion measure, Fidelity+ and the GraphFramEx characterisation score) select GuidedBP on MUTAG under a Bemis–Murcko scaffold split, whose agreement with the mutagenic nitro motif is GT AUROC 0.016, the worst of the 6 attributors audited, while the ground truth ranks GNNExpl. best at 0.671 (median paired gap 0.767 AUROC, Wilcoxon $p < 0.001$, Benjamini–Hochberg $q < 0.001$ over the 12 selection tests). The faithfulness–truth rank correlation is $+0.829$ in distribution and -0.600 under shift; on a synthetic set with *exact* node labels the same inversion appears with a smaller penalty. Faithfulness itself does not fall under shift (median change -0.001 across 43 cells audited on both splits), so a faithfulness metric gives no warning that its relationship to correctness has broken. Using 5158 committed per-molecule records we further find that ground-truth localisation falls *below chance* on confidently-wrong predictions (0.327, $n = 26$) while their faithfulness looks unremarkable, and that the calibration–reliability link reverses sign between a per-cell and a pooled analysis. Results are the committed `full.yaml` run: 86 of 88 cell-runs completed, 2 were skipped as unreachable, and 38 rows carried from an earlier run are labelled throughout and never merged into a this-run aggregate. Every number, figure and table regenerates from the committed artifacts.

Introduction

A medicinal chemist is shown a graph neural network's prediction that a candidate is hERG-active, together with a heat map over its atoms. The heat map concentrates on a piperidine nitrogen; the chemist finds that plausible, and the series is deprioritised. Nothing in that workflow ever established that the highlighted atoms are the atoms the model actually used, that they are the atoms that matter chemically, or that the highlighting behaves the same way on the next scaffold. Each of those is a separate question, and the answers come apart.

They come apart in a way that is measurable, and that gets worse in exactly the setting drug discovery operates in. An attribution can be *faithful*, in that the atoms it ranks highest really are the atoms whose removal moves the model's output most, and simultaneously *wrong*, in that those atoms are not the substructure that determines the property. Across the 46 committed cells in this study that carry a node-level ground truth, 11 are positively faithful to the model and yet *anti*-aligned with the true motif (ground-truth AUROC below the chance value of 0.5). A benchmark that scores only faithfulness rates all of them as fine.

In practice the problem takes the form of a selection problem. With no ground truth available, which is the situation on every real molecular dataset in this study, an attributor has to be chosen using a faithfulness or fidelity metric. On the two cell families that do have a ground truth and were audited on both splits, we can check whether that choice is the right one while changing *only* the split. Under a Bemis–Murcko scaffold split on MUTAG, all 3 ranking metrics (our occlusion measure, Fidelity+ and the GraphFramEx characterisation score alike) select GuidedBP, whose agreement with the mutagenic nitro motif is GT AUROC 0.016, while the ground truth ranks GNNExpl. best at 0.671. The median per-molecule gap is 0.767 AUROC (paired Wilcoxon $p < 0.001$, $n = 20$; Benjamini–Hochberg $q < 0.001$ across the 12 selection tests), and the rank correlation between faithfulness and correctness is *negative* (-0.600

to −0.657). In distribution on the same dataset, two of the three metrics instead select the ground-truth-best attributor and the rank correlation is +0.829.

None of this makes faithfulness metrics wrong. They answer a different question from the one a chemist is asking, and the gap between the two questions widens under distribution shift. Molecular property prediction is deployed under scaffold shift almost by construction, since the point of a screening model is to score series the training set does not contain, and scaffold-split evaluation is now standard practice for the *predictions* themselves [26, 27]. The evaluation protocol for the *explanations* has not followed: it does not stratify by that shift at all.

Contributions. This paper presents **MolSanity**, a reliability-audit framework for feature attributions on molecular GNNs. Rather than propose another attribution method, it wraps the canonical implementations (Captum [17], PyTorch Geometric [18], RDKit [19]) and audits them. Specifically:

- We define a six-axis audit over a motif-native (RDKit) decomposition: coherence, occlusion–attribution faithfulness, ground-truth localisation, cross-checkpoint stability, calibration linkage and confidence/correctness regime stratification, with field-standard Fidelity±, sparsity, the GraphFramEx characterisation score [10] and the PyG/DIG unfaithfulness metric [11] computed on the *same* molecules for direct comparability (The MolSanity audit).
- We show that a faithfulness-only ranking picks the wrong attributor under scaffold shift, in a design where only the split varies, so the effect is not confounded with a change of dataset (A fidelity-only ranking picks the wrong attributor under shift).
- We show reliability is a property of the cell, not of the attributor: the backbone ordering does not survive the change of split ($\rho = -0.400$ over 5 backbones on the exact-ground-truth set), and neither does the attributor ordering ($\rho = 0.486$ on MUTAG, 0.657 on SynthMotifs). Faithfulness itself does *not* shift systematically (across the 43 cells audited on both splits the median change is −0.001), so what shift breaks is the *relationship* between faithfulness and correctness, not the level of faithfulness.
- Using 5158 per-molecule audit records, we stratify reliability by confidence/correctness regime and test the calibration-linkage hypothesis directly. It holds per cell (median $\rho = 0.056$, 21 cells significantly positive vs. 10 negative) but *reverses* when cells are pooled (−0.273), a Simpson's-paradox trap for anyone reporting a single aggregate number.
- We report the negatives with the positives: cells whose ground truth is degenerate, a dataset that trains but yields no usable node labels, rows carried over from an earlier run (labelled, never merged into a this-run aggregate), and the 78 cells where the correctness axis simply does not exist and is marked as such rather than scored.

Related work

Attribution methods. The attributors audited here are the standard ones. The gradient family (Saliency [2], Input×Gradient, Guided Backpropagation [3] and Integrated Gradients [1]) is model-agnostic and cheap. Graph-specific methods learn a mask over the input: GNNExplainer [4] optimises a soft edge/feature mask per instance, PGExplainer [5] amortises that into a parametric mask network, and SubgraphX [6] searches connected subgraphs with Monte-Carlo tree search under a Shapley objective. Graph analogues of CAM/Grad-CAM were introduced by Pope et al. [7]; Yuan et al. [8] survey the space. In every case we call the published implementation rather than reimplementing it.

Evaluating explanations. Two evaluation traditions exist. *Faithfulness* measures whether the explanation tracks the model: Fidelity± and sparsity [8], the GraphFramEx characterisation score that combines them [10], and the unfaithfulness metric shipped with DIG/PyG [11]. *Correctness* measures whether the explanation recovers a known rationale, which requires ground-truth substructures, typically synthetic [9]. The tension between the two is known: Faber et al. [13] argue that comparing against ground truth is misleading when the trained model does not use that rationale, and Sanchez-Lengeling et al. [14] built the first systematic attribution benchmark for molecular GNNs on exactly this basis. Outside graphs, sanity checks [15] and retraining-based benchmarks [16] made a parallel case that plausible-looking attributions can be uninformative. MolSanity's position is that neither axis subsumes the other, and that what needs measuring is the *joint* behaviour of the two under distribution shift.

Frameworks, and the explicit delta. Four lines of work are close enough that a citation is not a sufficient answer. Table 1 scores all of them on the same capabilities; what follows states what each row of that table means.

Sanchez-Lengeling et al. [14] is the nearest neighbour and the earliest. It built the first systematic attribution benchmark for molecular GNNs on tasks whose ground-truth substructure is known by construction, establishing that attribution methods should be scored against a known rationale rather than against how plausible a chemist finds the highlighting. MolSanity inherits that premise. What it adds is the faithfulness axis measured on the *same* molecules and the joint behaviour of the two. A benchmark that reports correctness alone cannot show that a faithfulness-based choice of attributor is the wrong choice, because it never makes that choice. The selection-consistency test of The selection-consistency test makes it, and its outcome changes with the split.

MolFaith [12] is the closest work on the faithfulness axis and is larger than this study on that axis: eight attribution methods, two molecular representations and five architectures over roughly 14,000 molecules, against the 5158 per-molecule records here. We make no claim to broader faithfulness coverage. Two of its findings bear directly on ours. First, it reports that some methods are inherently more faithful than others largely independently of architecture. We find the same stability from the other direction: faithfulness does not shift systematically across the 43 cells audited on both splits (median change −0.001). Two independent studies at different scales agreeing that faithfulness is a comparatively stable property of the method is the premise our result then acts on, because the ordering that does *not* survive the change of split is the ground-truth one ($\rho = 0.486$ on MUTAG, 0.657 on SynthMotifs). A stable faithfulness ranking is therefore not evidence of a stable correctness ranking, which is what a practitioner with no rationale labels has to assume when they use faithfulness to choose a method. Second, it reports that faithfulness varies markedly per molecule; our regime stratification locates part of that variation, since ground-truth localisation collapses below chance precisely on confidently-wrong predictions while faithfulness on the same molecules looks unremarkable.

GraphXAI [9] and *GraphFramEx* [10] are general-graph frameworks: GraphXAI supplies generators with exact node ground truth and an evaluation library, GraphFramEx systematises fidelity-style evaluation across explainers and tasks. Both are broader than MolSanity across graph domains and narrower within chemistry. MolSanity scores chemically meaningful units (SSSR rings, Murcko scaffold, BRICS fragments) rather than individual atoms, and stratifies by the split that governs molecular deployment. *DIG* [11] is a library rather than a protocol; we wrap its unfaithfulness metric, and its SubgraphX implementation is the coverage gap recorded in Limitations and threats to validity.

What is new here, and what is not. Ground-truth localisation, fidelity metrics, motif decompositions and scaffold splits all exist already. Two rows of Table 1 we do not find in any of the four: cross-checkpoint stability and calibration linkage. The contribution we would defend is neither of those rows alone, but the result they were built to support, that the two evaluation traditions above invert under the shift molecular models are deployed into and that no metric on the faithfulness side signals it.

The MolSanity audit

Notation

A molecule is a graph $G = (V, E)$ with node features $X \in \mathbb{R}^{|V|\times d}$ and edge features $Z \in \mathbb{R}^{|E|\times d'}$. A trained model f_θ maps G to logits $f_\theta(G) \in \mathbb{R}^C$ (classification) or a scalar (regression); t indexes the explained output. An *attributor* is a map $A : (f_\theta, G, t) \mapsto a \in \mathbb{R}^{|V|}$ assigning a real score to every atom.

RDKit decomposes G into a motif cover $\mathcal{M}(G) = \{M_1, \ldots, M_K\}$, $M_k \subseteq V$, from SSSR rings, the Bemis–Murcko scaffold [31] and BRICS-style fragments. Writing $a_v^+ = \max(a_v, 0)$, the motif-level score is

$$s_k = \sum_{v\in M_k} a_v^+, \qquad k = 1, \ldots, K. \tag{1}$$

Working at motif rather than atom granularity is what makes the audit chemically legible: a chemist reasons about a nitro group, not about atom 14.

For $S \subseteq V$ let $G\backslash S$ denote G with the features of S zeroed through the model's node-mask multiplier, $X' = X \odot (\mathbf{1} - \mathbb{1}_S)$, leaving the topology intact.

The five audit statistics

(i) Faithfulness. Occlusion of motif M_k changes the explained output by

$$\Delta_k = f_{\theta,t}(G) - f_{\theta,t}(G\backslash M_k). \tag{2}$$

An attribution is faithful when it ranks motifs as occlusion does, measured by the Spearman rank correlation over the K motifs of one molecule,

$$\rho_{\text{occ}}(G) = \text{Spearman}\big((s_k)_{k\le K}, (\Delta_k)_{k\le K}\big) \in [-1, 1]. \tag{3}$$

$\rho_{\text{occ}} = 1$ means the attributor recovers the causal ordering exactly; $\rho_{\text{occ}} = 0$ means it is uninformative about it; and $\rho_{\text{occ}} < 0$ means it *inverts* it, which is strictly worse than an uninformative explanation. Undefined cases ($K < 2$, or zero variance in either argument) are recorded as missing, never as 0.

We also reproduce the field-standard fidelity pair [8]. With τ the 80th percentile of a^+ and $S_\tau = \{v : a_v^+ \geq \tau\}$

Table 1: Where the audit sits relative to existing evaluation frameworks. ✓ = core capability, ∼ = partial or possible but not central, ✗ = not a focus. The contribution is the combination, not any single row: ground-truth validation and faithfulness both exist elsewhere, but not jointly with cross-checkpoint stability, calibration linkage and shift stratification over a motif-native decomposition.

Capability	GraphXAI	GraphFramEx	DIG	MolFaith	MolSanity
Ground-truth explanation benchmarks	✓ (synthetic)	∼	✓	✗	✓ (synthetic + motif-proxy)
Faithfulness / fidelity metrics	∼	✓	✓	✓	✓ (reproduced for comparability)
Molecular-motif-native audit (RDKit)	✗	✗	∼	∼	✓ (SSSR + Murcko + BRICS)
Occlusion–attribution agreement	∼	✓	∼	✓	✓
Cross-checkpoint stability	✗	✗	✗	✗	✓
Calibration linkage	✗	✗	✗	✗	✓
Scaffold-shift regime stratification	✗	∼	✗	✗	✓ (the core differentiator)
Multiple GNN backbones (agnostic)	∼	✓	✓	∼	✓ (GINE/GCN/GAT/MPNN/AttentiveFP)
Paired statistics (Wilcoxon, bootstrap CI)	∼	∼	∼	∼	✓
Wraps canonical implementations (no re-impl)	✓	✓	✓	✓	✓ (Captum/PyG/RDKit)

the salient set, and $p_t(\cdot)$ the softmax probability of the explained class,

$$\mathrm{Fid}^+ = p_t(G) - p_t(G \setminus S_\tau), \tag{4}$$
$$\mathrm{Fid}^- = p_t(G) - p_t(G \setminus \bar{S_\tau}), \tag{5}$$

higher Fid^+ and lower Fid^- being better, with sparsity $1 - |S_\tau|/|V|$. The GraphFramEx characterisation score [10] combines them as the weighted harmonic mean

$$\mathrm{char} = \frac{(w_+ + w_-)\,\mathrm{Fid}^+\,(1-\mathrm{Fid}^-)}{w_-\mathrm{Fid}^+ + w_+(1-\mathrm{Fid}^-)}, \qquad w_+ = w_- = \tfrac{1}{2}, \tag{6}$$

undefined outside $\mathrm{Fid}^\pm \in [0,1]$. That constraint matters for the regression cells, where Δ is taken in output space and $\mathrm{Fid}^\pm$ leaves that range.

(ii) Correctness. Where a node-level ground-truth mask $g \in \{0,1\}^{|V|}$ exists, correctness is the ability of a to separate the true substructure from the rest, scored by the area under the ROC curve of the min–max normalised attribution against g,

$$\begin{aligned}\text{GT-AUROC}(G) = \Pr[\tilde{a}_u > \tilde{a}_v \,|\, g_u{=}1,\, g_v{=}0] \\ + \tfrac{1}{2}\Pr[\tilde{a}_u = \tilde{a}_v],\end{aligned} \tag{7}$$

with u, v drawn uniformly from the positive and negative atoms. Chance is $\frac{1}{2}$; values below $\frac{1}{2}$ mean the attribution is *anti-aligned*, ranking the true motif systematically *below* the rest. When g is degenerate ($\sum_v g_v \in \{0, |V|\}$) the quantity is undefined and reported as such.

(iii) Coherence. Concentration and connectedness of the attribution mass: the Gini coefficient of a^+, the fraction of mass in the top 20% of atoms, the largest-connected-component fraction of S_τ in the induced subgraph, and the motif top-1 share $\max_k s_k / \sum_k s_k$. These are descriptive; Cross-task reliability shows they carry little reliability signal.

(iv) Stability. With θ^{early} an intermediate checkpoint of the same training run and $s^{\text{early}}, s^{\text{final}}$ the corresponding motif scores (1),

$$\mathrm{stab}(G) = \mathrm{Spearman}(s^{\text{early}}, s^{\text{final}}). \tag{8}$$

An explanation that reorders motifs between two checkpoints of one run is a weak basis for a chemistry decision.

(v) Regime and calibration. Logits are temperature-scaled on the validation fold [32]; let $c(G) = \max_j p_j(G)$ be the calibrated confidence and $y(G) \in \{0,1\}$ the correctness of the prediction. With $\tau_c = 0.8$, each molecule is assigned

$$R(G) = \begin{cases} \text{conf-correct} & c \geq \tau_c,\ y = 1, \\ \text{conf-error} & c \geq \tau_c,\ y = 0, \\ \text{borderline} & c < \tau_c. \end{cases} \tag{9}$$

Calibration linkage is the within-cell rank correlation between per-molecule calibration quality and reliability,

$$\rho_{\text{cal}} = \mathrm{Spearman}(1 - |c - y|,\ \rho_{\text{occ}}), \tag{10}$$

positive values supporting the hypothesis that better-calibrated predictions carry more trustworthy attributions.

The selection-consistency test

The audit's central question is operational rather than descriptive. A practitioner with no ground truth must choose an attributor using a faithfulness statistic; we ask whether that choice agrees with the one the ground truth would make.

Fix a *cell* $c = (\mathcal{D}, f_\theta, \pi)$, meaning a dataset, a trained backbone and a split, together with a set $\mathcal{A}$ of attributors, each audited on the same molecule sample $\mathcal{G}_c$. For attributor A write $\bar{m}(A) = |\mathcal{G}_c|^{-1} \sum_{G \in \mathcal{G}_c} m_A(G)$ for the cell mean of statistic m. Define the ground-truth-optimal and the faithfulness-optimal attributor

$$A^\star = \arg\max_{A \in \mathcal{A}} \overline{\mathrm{GT}}(A), \qquad \hat{A}_m = \arg\max_{A \in \mathcal{A}} \bar{m}(A), \quad (11)$$

for each faithfulness statistic $m \in \{\rho_{\mathrm{occ}}, \mathrm{Fid}^+, \mathrm{char}\}$. Two quantities follow. The *selection gap* is the paired per-molecule difference in correctness between the two choices,

$$\delta_m(G) = \mathrm{GT}_{A^\star}(G) - \mathrm{GT}_{\hat{A}_m}(G), \qquad G \in \mathcal{G}_c, \quad (12)$$

tested against H_0 : median $\delta_m = 0$ with a two-sided Wilcoxon signed-rank test on the molecules both attributors audited. The *selection-consistency coefficient* is the rank correlation between the two orderings over $\mathcal{A}$,

$$\rho^{\mathrm{sel}}_m = \mathrm{Spearman}_{A \in \mathcal{A}}(\bar{m}(A), \overline{\mathrm{GT}}(A)) \in [-1, 1]. \quad (13)$$

$\rho^{\mathrm{sel}} \approx 1$ means faithfulness is a sound surrogate for correctness in that cell; $\rho^{\mathrm{sel}} \approx 0$ means it is uninformative; $\rho^{\mathrm{sel}} < 0$ means it is actively misleading, in that ranking by faithfulness does worse than ranking at random. Because $\mathcal{D}$, f_θ and $\mathcal{A}$ are held fixed and only π varies between the two arms of each comparison, any change in ρ^{sel} is attributable to the split rather than to a change of dataset or model.

Estimation and inference

Cell means are reported with 95% percentile bootstrap intervals over molecules (2000 resamples, fixed seed). Attributor comparisons within a cell are paired on molecule identity and tested with the two-sided Wilcoxon signed-rank test; correlations are Spearman throughout, as all statistics here are ordinal in nature and several are heavy-tailed. Sample sizes are the audited molecules per cell (20–120), which is small enough that we treat individual p-values as descriptive and report them unadjusted for multiplicity, leaning on replication across cells rather than on any single test. Where a statistic is undefined for a molecule it is dropped from that statistic's mean only, and the retained n is reported alongside.

Implementation

The pipeline is staged and idempotent: each stage writes a marker keyed to a content hash of its configuration, every cell is isolated so one failure cannot abort a sweep, and trained checkpoints are content-addressed so an audit reproduces without retraining. Seeds, library versions, git revision and hardware are recorded per run. Attributors are the canonical implementations [17, 18], wrapped behind one interface rather than reimplemented.

Experimental setup

Datasets. Tier-1 provides the ground-truth anchor: **SynthMotifs** and its larger twin **SynthMotifsXL**, generated offline (Barabási–Albert base plus house/cycle motifs) with *exact* node ground truth, and **MUTAG** [29, 30] with the nitro proxy. Tier-2 is MoleculeNet [27]: **BBBP** and **BACE** (classification), **ESOL**, **FreeSolv** and **Lipophilicity** (regression). Tier-3 covers chemistry-meets-medicine endpoints: **ClinTox**, **SIDER** and **Tox21** as single-task binary views of the MoleculeNet panels, and **DILI** (drug-induced liver injury) and **hERG** (cardiotoxicity) via the Therapeutics Data Commons [28]. TDC sets are featurised with the same PyG encoding as MoleculeNet so the backbones are drop-in. BA-2Motifs and ShapeGGen [9] are attempted and gracefully skipped: the former's download returns HTTP 403 in this environment, the latter's install depends on the pre-2.5 PyG compiled extensions. Credential-gated resources are out of scope and never worked around.

Backbones. GINE [20, 21], GCN [22], GAT [23], MPNN [24] and AttentiveFP [25], sharing a pooling head, calibration and splits.

Attributors. Integrated Gradients, Saliency, Guided Backpropagation and Input×Gradient through Captum [17]; GNNExplainer and PGExplainer through PyTorch Geometric [18]. SubgraphX [6] requires DIG, which installs but cannot import in this environment (no `torch_sparse` wheel at the pinned torch version), so it is left blocked rather than approximated. PGExplainer is classification-only by construction: its mask network trains against class logits, so it is absent from the regression cells.

Splits. The primary split is a deterministic, class-aware Bemis–Murcko scaffold split [31], the protocol used to evaluate molecular property models under realistic generalisation [26] (whole scaffold groups move together, so there is no leakage, but every fold is guaranteed to contain every class), with a random split as the in-distribution reference. Imbalanced classification uses inverse-frequency class weighting and AUC-based model selection. Every comparison in this paper that is labelled "shift" versus "in-distribution" is these two splits on the same dataset.

Table 2: The run ledger. Outcome of every cell the `full.yaml` sweep attempted, as recorded in `results/PROGRESS.md`. *carried* counts cells whose row still appears in the results matrix from an earlier reduced-budget CPU run because this run's attempt failed. Those rows are marked [c] throughout this paper and are never mixed into a this-run aggregate.

dataset	done	failed	skipped	carried
BA-2Motifs	2	0	0	0
BACE	4	0	0	3
BBBP	14	0	0	3
ClinTox	4	0	0	2
DILI	2	0	0	3
ESOL	8	0	0	3
FreeSolv	2	0	0	4
Lipophilicity	2	0	0	4
MUTAG	20	0	0	0
SIDER	4	0	0	3
ShapeGGen	0	0	2	0
SynthMotifs	20	0	0	0
Tox21	2	0	0	3
hERG	2	0	0	4
total	**86**	**0**	**2**	**38**

Failure reasons. .

Scale of the study. The sweep trains every (dataset, backbone) pair on both splits and audits every (dataset, backbone, attributor, split) combination, giving 86 audited combinations and 5158 per-molecule audit records in total. Audited samples are 20–120 molecules per combination, drawn from the held-out test fold. Exact software versions, hardware, seed and the per-combination outcome ledger are recorded with the released artifacts (Data and code availability).

Coverage. Of the 88 planned combinations, 86 completed and 2 could not be attempted because a dependency (ShapeGGen) is unavailable in this environment (Table 2).

A further 38 rows in the results matrix come from an earlier, reduced-budget run on different hardware: they cover attributor sweeps on the Tier-2/3 molecular sets that the primary configuration does not schedule, plus the SynthMotifsXL arm. These are genuine measurements, but they are not comparable in budget to the rest, so they are marked [c] throughout, excluded from every aggregate computed over per-molecule records, and excluded entirely from the analysis in A fidelity-only ranking picks the wrong attributor under shift. In total the matrix holds 101 classification and 23 regression rows, of which 46 carry a node-level ground truth.

Results

Faithfulness is not correctness

Table 6 lists every committed cell in which attribution *correctness* can be measured at all, and Figure 1 plots the joint distribution. Of the 46 cells with both metrics defined, 18 are anti-aligned with the ground truth (GT AUROC < 0.5) and 11 of those are simultaneously positively faithful to the model. That combination is the failure mode a faithfulness-only audit cannot see, because from the model's point of view nothing is wrong.

The clearest molecular instance is the MUTAG scaffold block of Table 6, where dataset, backbone and split are fixed and only the attributor varies. Guided Backpropagation, Saliency and Input×Gradient are the three *most* faithful attributors on that cell ($\rho = 0.701, 0.611, 0.640$) and the three *worst* at recovering the nitro motif (GT AUROC 0.016, 0.009, 0.049). GNNExplainer, which the ground truth ranks best at 0.671, is less faithful (0.607) than all three. On this cell the two axes are not merely uncorrelated; they are inverted.

A fidelity-only ranking picks the wrong attributor under shift

If faithfulness and correctness dissociate, then ranking attributors by a faithfulness metric, which is what an evaluation framework must do when no ground truth is available, can select the wrong method. Two cell families let us test this cleanly: each has a node-level ground truth, all 6 attributors audited on the same molecules, and both splits from this run, so the in-distribution and shift arms differ *only* in the split (Table 5, Figure 3).

MUTAG, under shift. The ground truth ranks GNExpl. best (0.671). All 3 metrics instead rank GuidedBP first, at 0.016, the worst of the 6. On paired per-molecule GT AUROC over the 20 shared molecules the median gap is 0.767 with $p < 0.001$, and $q < 0.001$ after controlling the false discovery rate over all 12 selection tests, so this contrast is not an artefact of testing many of them. The faithfulness–truth rank correlation is negative for every metric: -0.600 for occlusion ρ, -0.657 for Fidelity+ and -0.657 for characterisation. A framework ranking on faithfulness here fails to find the best attributor and systematically prefers the worst.

MUTAG, in distribution. The same experiment on the random split behaves differently. Occlusion ρ and Fidelity+ both select PGExpl., which is also the ground-truth-best, and their rank correlation with truth is $+0.829$. The characterisation score is the exception, selecting Input×Grad at 0.054 ($p < 0.001$) with rank

Table 3: Molecular regression cells. Occlusion faithfulness is computed in output space (predicted-value shift) rather than probability space. In 18 of the 23 committed regression cells the attribution–occlusion agreement is *negative*: the atoms an attributor ranks highest are not the atoms whose removal moves the prediction most. ESOL·GAT, Lipophilicity·GAT, Lipophilicity·GINE run positive. Note the Fidelity$\pm$ columns leave the $[-1, 1]$ range a probability-space fidelity would occupy (up to 5.01), which is why we report but do not interpret their sign. Bold marks the highest occlusion ρ per dataset (emphasis only). PGExplainer is classification-only and therefore absent by construction.

dataset	backbone	attributor	split	n	RMSE	MAE	R^2	motif top-1	occ. ρ	occ. top-1	Fid+	stab.
ESOL	GAT	IG	random	100	0.730	0.549	0.888	0.850	0.845	0.970	3.52	0.908
ESOL	GAT	IG	scaffold	100	0.793	0.627	0.838	0.838	**0.862**	0.960	5.01	0.935
ESOL	GCN	IG	random	100	0.951	0.746	0.809	0.870	−0.534	0.330	−0.83	0.936
ESOL	GCN	IG	scaffold	100	1.017	0.793	0.734	0.870	−0.509	0.350	−0.88	0.965
ESOL	GINE	GNNExpl.	random	100	0.788	0.599	0.869	0.868	−0.873	0.230	−1.35	0.948
ESOL	GINE	GNNExpl.	scaffold	100	0.929	0.724	0.778	0.853	−0.739	0.280	−0.91	0.974
ESOL[c]	GINE	GuidedBP	scaffold	20	1.132	0.918	0.671	0.659	−0.741	0.150	−0.58	—
ESOL[c]	GINE	Input×Grad	scaffold	20	1.132	0.918	0.671	0.881	−0.793	0.100	−1.02	—
ESOL	GINE	IG	random	100	0.788	0.599	0.869	0.878	−0.944	0.230	−1.73	0.960
ESOL	GINE	IG	scaffold	100	0.929	0.724	0.778	0.878	−0.778	0.290	−1.39	0.954
ESOL[c]	GINE	Saliency	scaffold	20	1.132	0.918	0.671	0.878	−0.798	0.100	−1.00	—
FreeSolv[c]	GCN	IG	scaffold	20	2.567	2.048	0.538	0.782	−0.434	0.400	−0.05	—
FreeSolv[c]	GINE	GNNExpl.	scaffold	20	2.347	1.733	0.614	0.766	−0.600	0.350	−0.42	—
FreeSolv[c]	GINE	Input×Grad	scaffold	20	2.347	1.733	0.614	0.805	**−0.370**	0.450	−0.36	—
FreeSolv	GINE	IG	random	65	1.486	1.081	0.803	0.864	−0.418	0.523	−0.66	0.847
FreeSolv	GINE	IG	scaffold	65	1.304	0.965	0.881	0.872	−0.589	0.446	−0.50	0.877
FreeSolv[c]	GINE	Saliency	scaffold	20	2.347	1.733	0.614	0.789	−0.425	0.450	−0.37	—
Lipophilicity[c]	GAT	IG	scaffold	20	0.940	0.746	0.391	0.762	0.458	0.900	−0.14	—
Lipophilicity[c]	GINE	GNNExpl.	scaffold	20	0.932	0.740	0.402	0.845	−0.499	0.000	−0.25	—
Lipophilicity[c]	GINE	Input×Grad	scaffold	20	0.932	0.740	0.402	0.699	−0.439	0.000	−0.28	—
Lipophilicity	GINE	IG	random	100	0.737	0.571	0.617	0.775	0.464	0.720	0.52	0.832
Lipophilicity	GINE	IG	scaffold	100	0.749	0.584	0.614	0.760	**0.575**	0.860	1.03	0.863
Lipophilicity[c]	GINE	Saliency	scaffold	20	0.932	0.740	0.402	0.705	−0.469	0.000	−0.30	—

correlation −0.829. The sign flip between the two arms is large (+0.829 to −0.600 for occlusion ρ), though this does not make in-distribution evaluation safe, only much less dangerous.

The exact-ground-truth arm agrees, more mildly. On SynthMotifs, where node labels are exact rather than a motif proxy, the same pattern appears with a smaller penalty. In distribution, occlusion ρ picks the ground-truth-best and the two field-standard metrics pick a neighbour whose GT AUROC is statistically indistinguishable from it (0.999 vs 1.000, $p = 0.109$). Under shift all three select IG (0.901) over the ground-truth-best Saliency (0.975), significantly so ($p < 0.001$), though here the rank correlations stay positive (0.429 to 0.657). The direction replicates across an exact and a proxy ground truth; the severity does not.

Two caveats. First, our own faithfulness metric fails alongside the field-standard ones on the MUTAG shift arm. We therefore do not claim that MolSanity's occlusion measure is a better selector. Under shift, *no* faithfulness metric in this experiment carries reliable information about which attributor is correct, which argues for measuring correctness rather than for building a better proxy. Second, 2 of the ground-truth cells are degenerate in a way that matters here: every audited molecule returns an identical GT AUROC of 1.000, including MUTAG·GINE·PGExpl., random ($n = 20$). That cell is the one the in-distribution MUTAG arm names as ground-truth-best, so the "no mismatch" verdict there rests on a cell with zero between-molecule variance and should not be read as a positive result. The shift arm, which carries the argument, does not involve it.

Reliability is a property of the cell

Figure 4 lays the two axes over the audited grid, and two further comparisons show that neither axis is a stable property of the attributor or of the architecture.

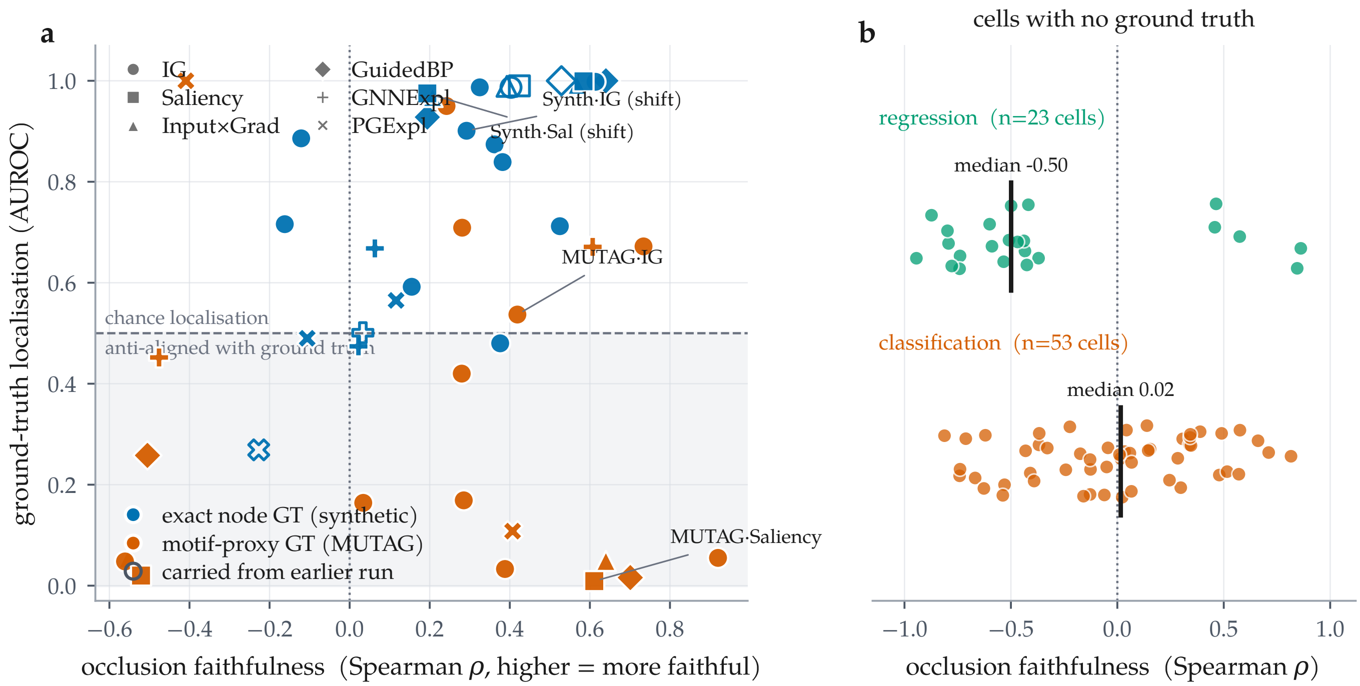

Figure 1: The two axes come apart, and for most cells only one of them exists. (**a**) Every committed cell carrying a node-level ground truth ($n = 46$), plotted as occlusion faithfulness against ground-truth localisation; colour distinguishes exact synthetic ground truth from the MUTAG motif proxy, marker distinguishes attributor, and open markers are rows carried from an earlier run. The shaded band is GT AUROC < 0.5: attributions there are anti-aligned with the true motif. 11 cells sit in the lower-right region, positively faithful to the model and anti-aligned with the truth. (**b**) The remaining 78 combinations, on real molecular datasets, have no published per-atom labels: only the horizontal axis is measurable for them.

Backbones. Holding the attributor fixed at Integrated Gradients and varying the backbone (Figure 5) gives a spread of 0.518 AUROC in distribution (GINE 0.998 down to GCN 0.480) and 0.395 under shift (GCN 0.987 down to GAT 0.592), comparable to the spread across attributors on the same dataset. The two orderings are negatively correlated ($\rho = -0.400$, 5 backbones): the architecture that localises the motif best in distribution is not the one that does so under shift. The attributor ordering is somewhat more robust across the same change of split ($\rho = 0.657$ over 6 attributors) but far from preserved.

The shift itself does not lower faithfulness. It would be tidy if scaffold shift simply made attributions less faithful. It does not. Across the 43 cells this run audited on both splits, occlusion faithfulness falls under the scaffold split in 22 of them, roughly half, with a median change of -0.001 (Figure 6b). We reported the opposite trend in an earlier, partial version of this matrix; with the grid complete it does not hold. What shift damages is the *relationship* between faithfulness and correctness, not the level of faithfulness, which is why a faithfulness metric gives no warning that anything has changed.

Where attributions fail: regimes and calibration

With per-molecule records available, two audit axes that previous drafts could only define are now measurable (Figure 7).

Regime stratification. Pooling the 5158 audited molecules by regime gives the audit's most directly actionable result. Ground-truth localisation is highest on confident-correct molecules (0.695, $n = 432$), drops to 0.409 on borderline ones ($n = 342$), and falls *below chance*, to 0.327, on *confident-error* molecules ($n = 26$). When the model is confidently wrong, its attribution points away from the substructure that matters rather than towards it. That is the regime a deployment cares most about and the one in which an explanation is most likely to be believed. Occlusion faithfulness gives no warning: it is 0.028 on those same confident-error molecules, no worse than the -0.201 of the confident-correct ones. The confident-error ground-truth estimate rests on 26 molecules and is reported with that n throughout; it is suggestive rather than precise.

Calibration linkage. The framework's hypothesis is that better-calibrated predictions carry more trustworthy attributions. Computed per cell it is supported on balance (median $\rho = 0.056$ across 74 classification cells,

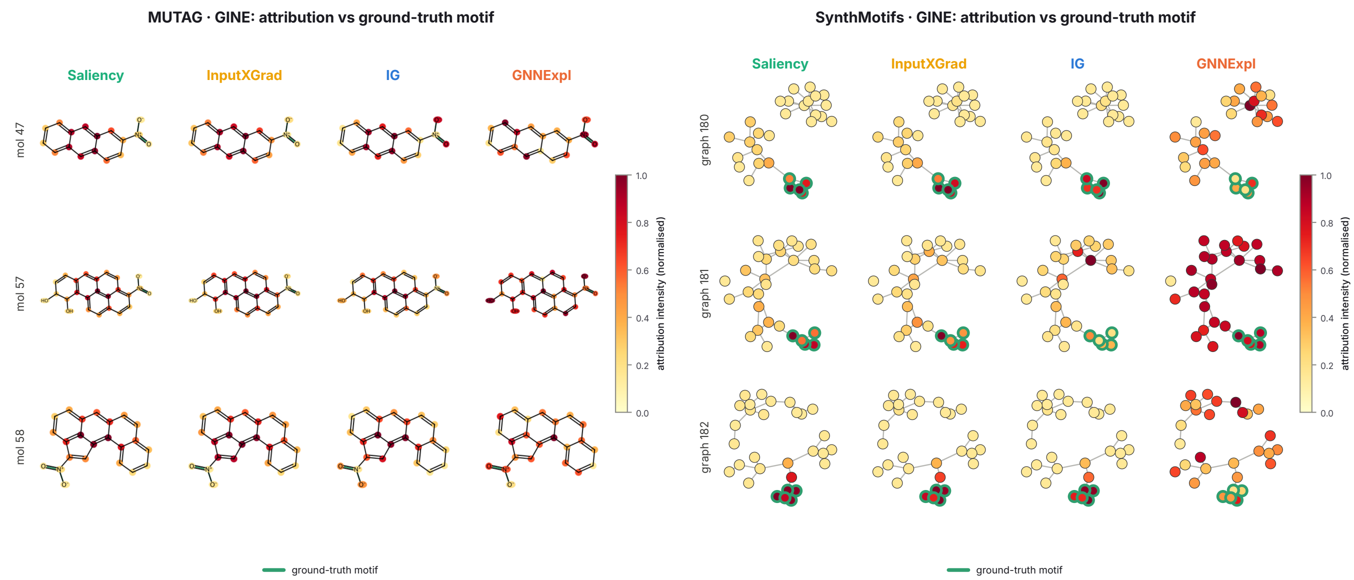

Figure 2: What the disagreement looks like on the structures themselves. Attribution intensity overlaid on the audited graphs for four attributors on the same molecules, rendered by the audit run: RDKit skeletal structures for MUTAG (left) and node-link diagrams for the synthetic set (right), with the ground-truth motif outlined in green. The quantitative gap of Table 5 is visible here as a qualitative one: on the synthetic graphs the gradient attributors concentrate on the outlined motif while GNNExplainer spreads mass across the Barabási–Albert base, and on MUTAG the gradient methods saturate the fused aromatic system rather than the nitro groups that carry the mutagenicity label. These panels are the pipeline's own output, reproduced unmodified.

21 significantly positive against 10 significantly negative), but the effect is cell-specific rather than universal. Pooling every molecule into a single correlation instead yields -0.273: the opposite sign. The pooled number is an artefact of mixing cells with different confidence distributions, and we report it only to warn against it.

Cross-task reliability

Tables 3, 7 and 8 give every audited combination on a real molecular dataset. Three cross-task patterns hold, and one apparent pattern turns out to be an artefact.

The regression faithfulness operator was mis-specified, and we withdraw its numbers. Across the 23 regression combinations, 18 have negative occlusion faithfulness (median -0.499), on cells whose predictive performance is well behaved (R^2 from 0.671 to 0.888 on ESOL). We do not interpret that pattern, and on inspection it is an artefact of our own operator rather than a property of the attributions. The operator compared two different quantities: the attribution was clipped at zero, keeping only atoms that push the prediction *up*, while the occlusion effect kept its sign. A motif that drives a solubility prediction strongly down therefore scored as unimportant on the attribution side and dominant on the occlusion side, which is sufficient on its own to produce a systematic negative correlation. The unbounded output-space Fidelity$\pm$ values, which reach 5.01 against the $[-1, 1]$ a probability-space fidelity occupies, are the visible symptom of the same mismatch. The corrected operator ranks by magnitude on both sides, reports the GraphFramEx characterisation score as undefined rather than clipping a standard-deviation-space shift into $[0, 1]$, and adds a bounded statistic giving the share of the total occlusion effect carried by the salient atoms. It is implemented and tested but postdates this sweep, so Table 3 is retained for completeness and the regression cells are excluded from every faithfulness claim in this paper. The classification path, on which every other result rests, is unchanged.

Coherence is high nearly everywhere, and says little. Motif top-1 share lies between 0.504 and 0.999 across the 96 molecular cells, with 73 of them inside a narrow 0.6–0.9 band, largely independent of whether the same cell is faithful or anti-faithful. Concentration on a single motif is easy to achieve and, on this evidence, close to uninformative about reliability. It belongs in the audit as a descriptive statistic, not as a quality score.

Stability is high but not uniform. Median cross-checkpoint stability over 86 cells is 0.833, with a minimum of -0.014, a cell whose early and final attributions are essentially uncorrelated. By attributor, GNNExplainer is the most stable (median 0.915) and Integrated Gradients sits at 0.845 over 58 cells; the per-attributor counts are very unequal in this run, so these medians are descriptive rather than a controlled comparison.

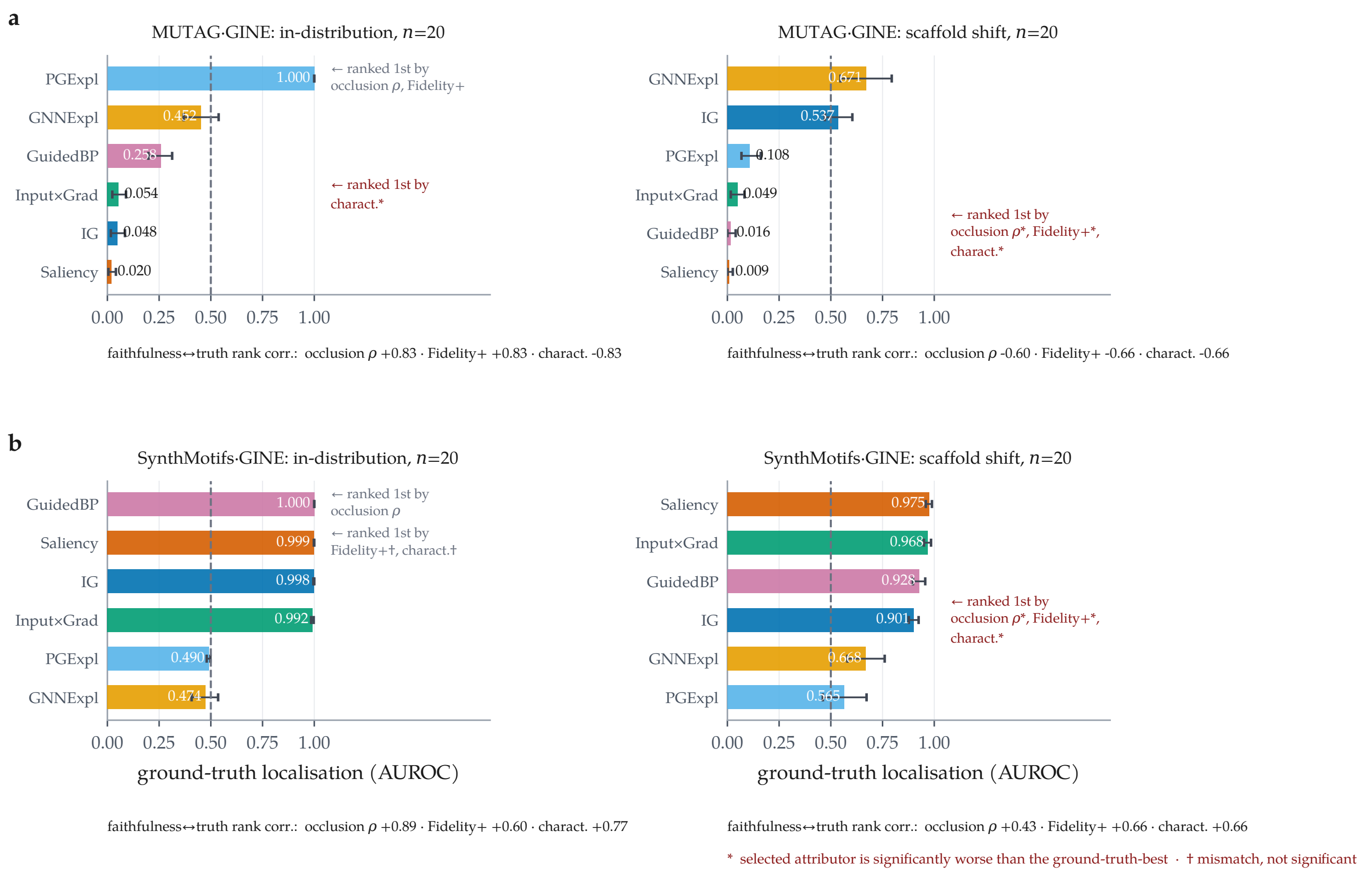

Figure 3: Under scaffold shift a faithfulness-only ranking selects the wrong attributor; in distribution it mostly does not. Ground-truth localisation per attributor, with bootstrap 95% confidence intervals over molecules. Within each row the dataset, backbone and attributor set are fixed and only the split changes. (**a**) MUTAG (motif-proxy ground truth): in distribution two of three metrics select the ground-truth-best attributor; under shift all three select GuidedBP, the worst-localising of the six, significantly so (marked *). (**b**) SynthMotifs (exact node ground truth): the same direction with a smaller penalty. In distribution the mismatched picks are statistically indistinguishable from the best (marked †); under shift they are not. Rank correlations between each faithfulness metric and ground truth are given below each panel.

Table 4 gives the paired attributor contrasts behind Section A fidelity-only ranking picks the wrong attributor under shift, computed from the per-molecule records.

Negative and degenerate results

We record what did not work alongside what did.

Ground truth that was not being read. BA-2Motifs is a synthetic benchmark whose whole purpose is to carry exact node labels. It trains in this run and produces 2×100 audited molecules, but both its cells sit in the no-ground-truth block, because the loader found no node labels to read. The cause was ours rather than the dataset's: PyG's `BA2MotifDataset` exposes only node features, an edge index and a graph-level label, and our extractor looked for a per-node field that is never populated. The labels are recoverable from the released node ordering, in which the injected motif is appended after the Barabási–Albert base. That recovery is now implemented and verified structurally, by requiring the induced subgraph on those nodes to be a house or a five-cycle before the mask is used, and cross-checked against a generator that produces the same construction with independent labels. It is refused rather than guessed when the check fails, so a reordered loader degrades to the behaviour reported here. The numbers in this paper predate that fix: BA-2Motifs is counted as audited and not as a ground-truth cell throughout, and the third exact-ground-truth arm it will provide is not claimed here.

Degenerate ground truth. 2 ground-truth cells return the identical value 1.000 on every audited molecule, including MUTAG·GINE·PGExpl., random ($n = 20$). A cell with zero between-molecule variance cannot support a confidence interval or a paired test, and one of the two is the attributor the in-distribution MUTAG arm names as ground-truth-best. We report those cells with the rest and mark what rests on them.

The best model has the worst attributions. The highest-AUC molecular classification cell in the run

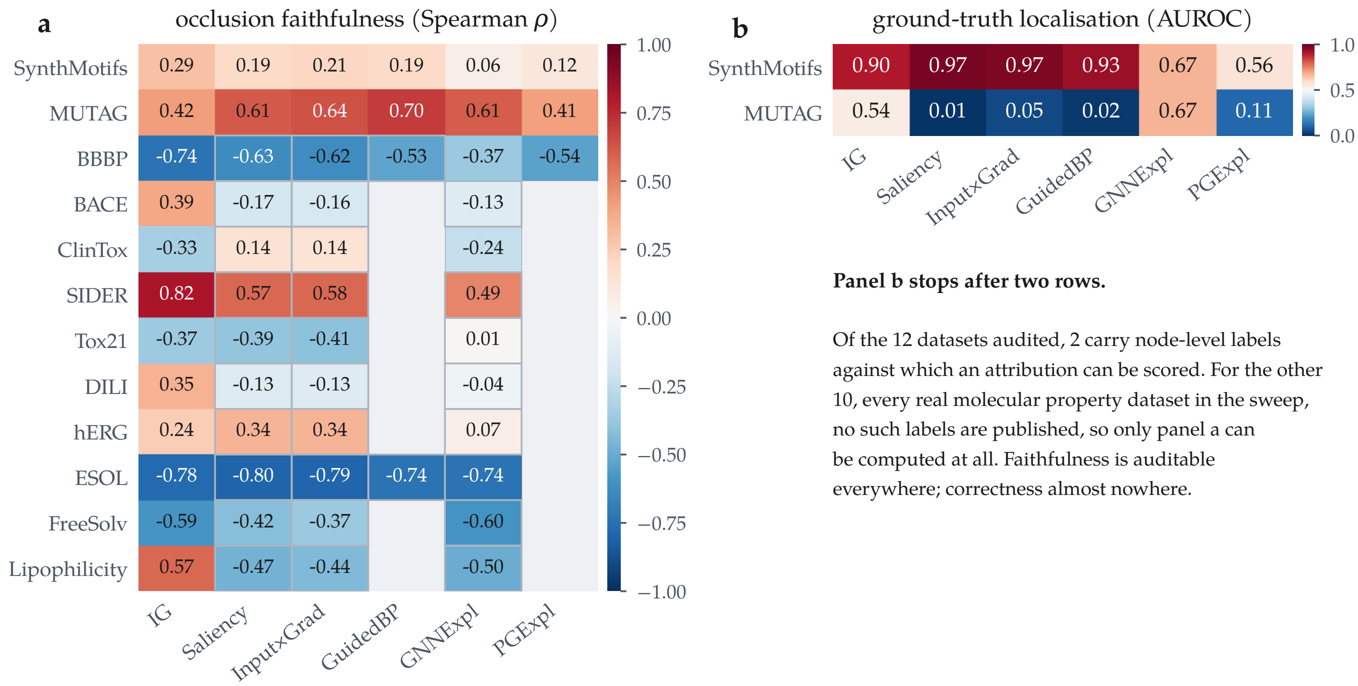

Figure 4: Reliability is a property of the (attributor, dataset) pair, not of the attributor. All committed GINE cells at the scaffold split. (**a**) Occlusion faithfulness over every audited dataset; blue = anti-faithful. (**b**) Ground-truth localisation, which exists for two datasets only. Panel b is short because correctness is unmeasurable elsewhere, not because it scores badly there: no molecular dataset in the sweep publishes per-atom labels. Blank cells were not audited; a thin outline marks a value carried from an earlier run.

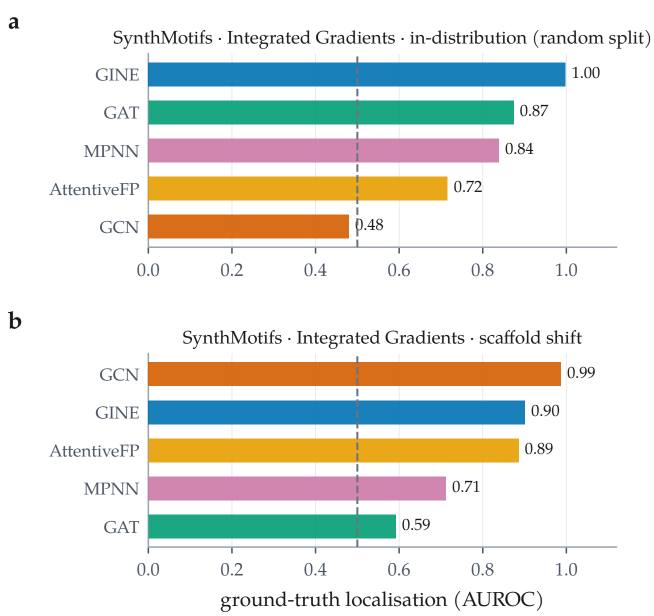

Figure 5: The backbone ordering does not survive the change of split. Integrated Gradients held fixed; ground-truth localisation across the 5 backbones on SynthMotifs, in distribution (**a**) and under scaffold shift (**b**). Dashed line is chance. Spearman correlation between the two orderings is −0.400 over 5 backbones.

is BBBP · GCN under the scaffold split, at accuracy 0.980, ROC-AUC 0.966 and ECE 0.013, so accurate and well calibrated. Its attributions are the most *anti*-faithful of any molecular cell in the matrix ($\rho = -0.812$): Integrated Gradients ranks the motifs of these molecules in close to the reverse of their causal order. Predictive quality, calibration and explanation reliability are three different axes.

A weak model can be faithfully explained. The converse also appears. The most faithful molecular cell is MUTAG · AttentiveFP ($\rho = 0.920$) on a model with accuracy 0.600 and ROC-AUC 0.958, barely better than chance. Faithfulness certifies that the explanation describes the model; it says nothing about whether the model is worth describing. Any reliability report that omits the predictive metrics can be satisfied by this cell.

Accuracy is the wrong headline on imbalanced sets. 38 committed rows report test accuracy $\geq$ 0.95, with ROC-AUC as low as 0.805. On Tox21, accuracy runs 0.960–1.000 against AUC 0.822–0.849. We report AUC alongside accuracy throughout for this reason, and the lowest-AUC cell in the run (BACE · GCN, AUC 0.507 at accuracy 0.850) is exactly the case where accuracy alone would mislead.

Blocked, and re-examined. The results reported here come from a run in which SubgraphX (via DIG) and ShapeGGen (via GraphXAI) were both recorded as uninstallable, on the grounds that they need the com-

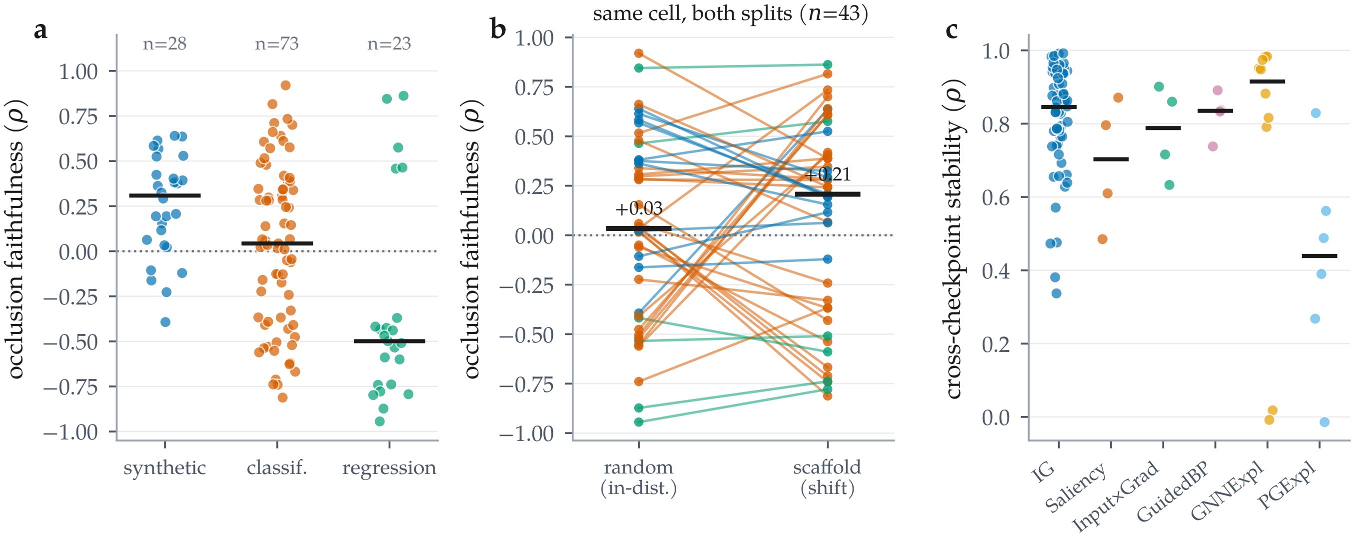

Figure 6: Distributions across all audited combinations. (**a**) Occlusion faithfulness by dataset regime; bars are medians. (**b**) The same cell on both splits, joined: faithfulness falls under the scaffold split in only 22 of 43 cells (median change −0.001), so shift does not systematically reduce faithfulness. (**c**) Cross-checkpoint stability by attributor; the mask-learning methods are not uniformly more stable than the gradient ones.

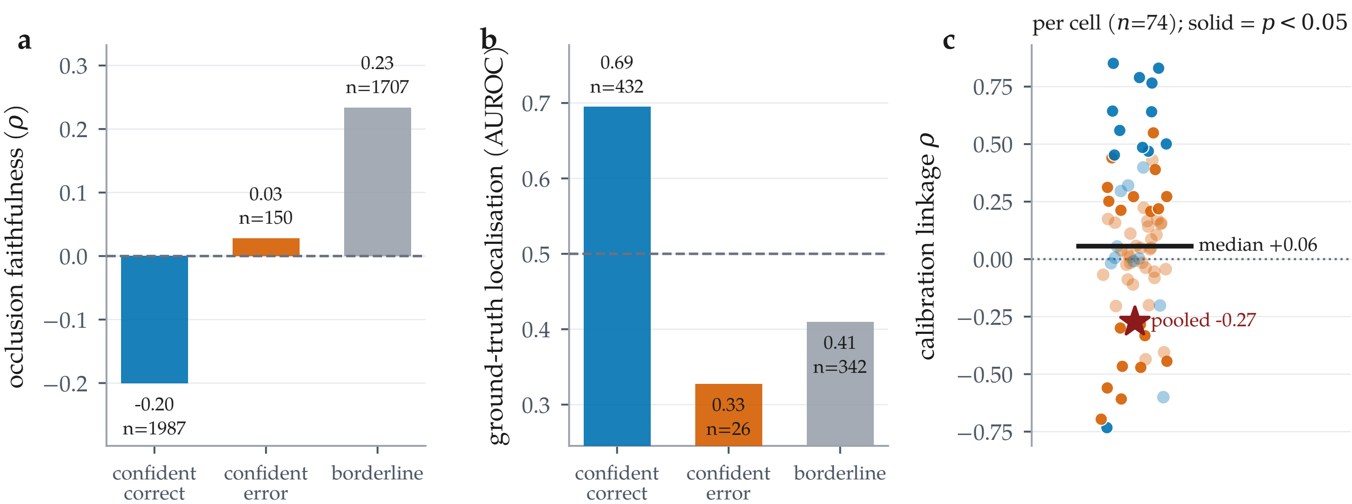

Figure 7: Regime stratification and calibration linkage, over the 5158 per-molecule audit records. (**a**) Occlusion faithfulness and (**b**) ground-truth localisation by confidence/correctness regime; n is printed per bar because ground truth exists for only a minority of molecules. Localisation on confident-*error* molecules falls below chance (0.327, n = 26) while occlusion faithfulness on the same molecules is unremarkable. (**c**) Calibration linkage computed per cell (solid = $p < 0.05$): the median cell is positive, but pooling all molecules into one correlation reverses the sign, a Simpson's paradox and a caution against reporting the pooled number.

piled `torch_sparse/torch_scatter` extensions for which no wheel exists at the pinned torch version. Revisiting that diagnosis, it was wrong in both cases. Those extensions build from source, DIG then installs, and SubgraphX is now wrapped and covered by tests in which it recovers a planted five-node motif exactly; it is enabled in the configuration and will populate the matrix on the next sweep. GraphXAI installs from a source checkout rather than a release, and ShapeGGen builds. It remains unintegrated for a different and more substantial reason: ShapeGGen is a node-classification benchmark on a single large graph, whereas every axis of this audit is defined per molecule at the graph level, so adopting it means extending the audit to node-level tasks rather than resolving a dependency. Both are recorded in the ledger, occupy no row in any table, and enter no aggregate. We report the earlier diagnosis alongside the correction because a coverage gap attributed to the environment turned out to be a gap in how hard we looked.

Table 4: Paired attributor comparisons on shared molecules, computed from the committed per-molecule records (Wilcoxon signed-rank on per-molecule occlusion faithfulness, $\Delta = A - B$). Same cell family as Table 5. p is the raw two-sided Wilcoxon value and q its Benjamini–Hochberg adjustment, controlling the false discovery rate over the contrasts within each cell block, which is the family a reader compares across.

A vs. B	n	median Δ	p	q
MUTAG, GINE, random split				
IG vs. Saliency	20	0.000	0.161	0.385
IG vs. Input×Grad	20	0.000	0.180	0.385
IG vs. GuidedBP	20	0.000	0.374	0.454
IG vs. GNNExpl.	20	−0.029	0.088	0.385
IG vs. PGExpl.	20	−0.086	0.108	0.385
Saliency vs. Input×Grad	20	0.000	0.236	0.409
Saliency vs. GuidedBP	20	0.000	0.600	0.600
Saliency vs. GNNExpl.	20	0.000	0.363	0.454
Saliency vs. PGExpl.	20	−0.186	0.083	0.385
Input×Grad vs. GuidedBP	20	0.000	0.533	0.571
Input×Grad vs. GNNExpl.	20	−0.029	0.134	0.385
Input×Grad vs. PGExpl.	20	−0.086	0.116	0.385
GuidedBP vs. GNNExpl.	20	−0.029	0.393	0.454
GuidedBP vs. PGExpl.	20	−0.114	0.245	0.409
GNNExpl. vs. PGExpl.	20	0.000	0.363	0.454
MUTAG, GINE, scaffold split				
IG vs. Saliency	20	−0.200	0.011	0.030
IG vs. Input×Grad	20	−0.250	0.003	0.030
IG vs. GuidedBP	20	−0.350	0.004	0.030
IG vs. GNNExpl.	20	−0.104	0.010	0.030
IG vs. PGExpl.	19	−0.036	0.527	0.609
Saliency vs. Input×Grad	20	0.000	0.292	0.365
Saliency vs. GuidedBP	20	0.000	0.027	0.051
Saliency vs. GNNExpl.	20	−0.029	0.962	0.962
Saliency vs. PGExpl.	19	0.143	0.047	0.078
Input×Grad vs. GuidedBP	20	0.000	0.063	0.094
Input×Grad vs. GNNExpl.	20	0.000	0.571	0.612
Input×Grad vs. PGExpl.	19	0.143	0.021	0.046
GuidedBP vs. GNNExpl.	20	0.050	0.276	0.365
GuidedBP vs. PGExpl.	19	0.483	0.012	0.030
GNNExpl. vs. PGExpl.	19	0.200	0.011	0.030

Discussion

What the audit buys. The single most useful thing MolSanity produces is not a score but a *negative certificate*. For 78 of the 124 committed rows, covering every real molecular dataset in the sweep, attribution correctness cannot be measured at all, because no per-atom explanation labels exist. Those cells can be audited for faithfulness, coherence, stability and calibration, and this paper's central result is that under shift none of those is a reliable stand-in for correctness. Making that gap explicit is more useful than filling it with a metric that looks like correctness and is not.

How attributions should be reported. Four practices follow directly from the matrix. (i) Report the regime: an attributor selected in distribution carries no warrant out of it, and on the one cell family where we can check, the selection is significantly wrong under shift. (ii) Report the backbone: the backbone ordering does not survive the change of split ($\rho = -0.400$), so "IG on a GNN" is under-specified. (iii) Report the predictive metrics next to the explanation metrics, because the accurate, well-calibrated BBBP · GCN cell has the most anti-faithful attributions in the matrix. (iv) Do not pool: the calibration-linkage correlation changes sign between the per-cell and the pooled analysis.

Why reliability is regime-, backbone- and dataset-dependent. A faithfulness metric asks about the model; a ground-truth metric asks about the chemistry; they agree only when the model has learned the chemistry. That coincidence is engineered into synthetic benchmarks and is not guaranteed on molecular data, particularly out of distribution, which is the situation Faber et al. [13] identify as making ground-truth comparison hazardous in general. Attribution evaluation inherits the model's inductive limitations, so the correct object of study is the cell, (dataset, backbone, attributor, split), and not the attributor alone.

Limitations and threats to validity

Ground truth is scarce, and one arm of it is a proxy. The two cell families that carry the argument have $n = 20$ audited molecules per arm. MUTAG, the molecular one, uses a chemically motivated nitro/aromatic-nitro SMARTS proxy rather than annotator labels: it encodes an assumption about which substructure ought to matter, and the trained model may use a different, equally valid rationale [13]. SynthMotifs, where the labels are exact, is a structural non-molecular graph task. The inversion replicates across both, which is the strongest available evidence, but neither arm is simultaneously molecular and exactly labelled. No public dataset is, which is the gap this paper is about.

Degenerate ground-truth cells. 2 cells return an identical GT AUROC on every audited molecule, including MUTAG·GINE·PGExpl., random (1.000, $n = 20$). Zero between-molecule variance means the bootstrap interval is uninformative and any "best attributor" verdict resting on such a cell is fragile; we flag them rather than drop them, and the shift arms that carry the argument do not involve them.

A dataset that trains but was not scored. BA-2Motifs loads and trains in this run, but its cells contribute faith-

Table 5: Does a faithfulness-only ranking pick the ground-truth-best attributor? Two cell families, each audited on both splits with the same backbone and the same six attributors, so within a family only the split changes and the contrast isolates distribution shift rather than confounding it with a change of dataset. Each row ranks the attributors by one faithfulness/fidelity metric and asks whether its top choice is the one the ground truth ranks best. p is a paired Wilcoxon test on per-molecule GT AUROC between the selected and the ground-truth-best attributor, over the molecules both audited, and q its Benjamini–Hochberg adjustment over all 12 selection tests in this table.

cell	regime	ranking metric	its top pick	pick GT	GT-best	GT-best	mismatch	Wilcoxon p	q	ρ(faith,GT)
MUTAG·GINE, n=20	in-distribution	occlusion ρ	PGExpl.	1.000	PGExpl.	1.000	no	—	—	0.829
		Fidelity+	PGExpl.	1.000	PGExpl.	1.000	no	—	—	0.829
		characterisation	Input×Grad	0.054	PGExpl.	1.000	**yes**	<0.001	<0.001	−0.829
MUTAG·GINE, n=20	scaffold shift	occlusion ρ	GuidedBP	0.016	GNNExpl.	0.671	**yes**	<0.001	<0.001	−0.600
		Fidelity+	GuidedBP	0.016	GNNExpl.	0.671	**yes**	<0.001	<0.001	−0.657
		characterisation	GuidedBP	0.016	GNNExpl.	0.671	**yes**	<0.001	<0.001	−0.657
SynthMotifs·GINE, n=20	in-distribution	occlusion ρ	GuidedBP	1.000	GuidedBP	1.000	no	—	—	0.886
		Fidelity+	Saliency	0.999	GuidedBP	1.000	**yes**	0.109	0.109	0.600
		characterisation	Saliency	0.999	GuidedBP	1.000	**yes**	0.109	0.109	0.771
SynthMotifs·GINE, n=20	scaffold shift	occlusion ρ	IG	0.901	Saliency	0.975	**yes**	<0.001	<0.001	0.429
		Fidelity+	IG	0.901	Saliency	0.975	**yes**	<0.001	<0.001	0.657
		characterisation	IG	0.901	Saliency	0.975	**yes**	<0.001	<0.001	0.657

fulness only and appear in the no-ground-truth block despite being a synthetic benchmark designed to have exact labels, because our extractor did not read the ordering its node labels are encoded in. The recovery is implemented and tested but postdates this sweep, so the third exact-ground-truth arm is available to the next run and is not part of any number reported here.

Single split, single seed. Each cell is one deterministic split at one seed. Cross-checkpoint stability partially probes run-to-run variability, but no confidence interval in this paper covers split or seed variance, and the per-cell molecule counts (20 on the synthetic arms) make individual contrasts underpowered.

Occlusion as a counterfactual. Zeroing node features takes the graph off the data manifold; a motif whose removal barely moves the output may be redundant rather than unimportant. This bears most heavily on the regression cells, where the output-space Fidelity$\pm$ values leave the range a probability-space fidelity occupies and we decline to interpret the sign.

Proxy ground truth. MUTAG's mask is a nitro/aromatic-nitro SMARTS proxy, not annotator labels. It encodes an assumption about which substructure ought to matter, and the trained model may use a different, equally valid rationale [13].

Coverage of attributors. SubgraphX contributes no row to this run and PGExplainer failed in it, so every conclusion about the perturbation family rests on GNNExplainer alone in the current-run cells. SubgraphX is wrapped and tested but its cells postdate this sweep, so the gap stands for the numbers reported here.

Multiplicity. The selection tests carry Benjamini–Hochberg adjusted q values over the family of 12 tests, and the attributor contrasts carry them over the family within each cell block, so the headline mismatches are false-discovery-rate controlled rather than descriptive. What the adjustment cannot repair is power: at $n = 20$ molecules per arm a contrast that fails to reach significance is weak evidence of no difference, not evidence of none.

Conclusion

Molecular GNN attributions are audited today for whether they describe the model. Across the committed audit matrix we find that this is not the same question as whether they describe the chemistry, and that the two diverge in the regime molecular property prediction actually operates in. On the one cell family where exact ground truth, several attributors and both splits coexist, a faithfulness-only ranking is harmless in distribution and selects a significantly worse attributor under scaffold shift. MolSanity's own faithfulness metric is no exception. Reliability does not transfer across backbones or across splits, and for the 78 molecular rows in the matrix, correctness cannot be measured at all with the labels the field currently has. The audit is what makes those failures visible; the pipeline, the figures and every number here regenerate from the committed artifacts as the grid fills in.

Table 6: Every committed cell in which attribution *correctness* is measurable. Ground truth is exact for the synthetic sets and a chemically motivated nitro-motif proxy for MUTAG. GT AUROC = 0.5 is chance; below 0.5 the attribution is anti-aligned with the true motif. Occlusion ρ is the attribution–occlusion rank agreement (higher = more faithful to the model). Rows marked [c] are *carried*: they survive in the results matrix from an earlier reduced-budget CPU run because the corresponding cell failed in the latest run (Table 2); every other row was produced by the run in Table 2. Bold marks the best GT AUROC and the best occlusion ρ within each dataset×split block (emphasis only; values are unaltered).

dataset	backbone	attributor	split	n	acc	AUC	GT AUROC	GT AUPRC	occ. ρ	Fid+	Fid–	stab.
MUTAG	AttentiveFP	IG	random	20	0.600	0.958	0.055	0.183	**0.920**	0.009	0.019	0.876
MUTAG	GAT	IG	random	20	0.550	0.583	0.709	0.627	0.281	0.307	0.338	0.966
MUTAG	GCN	IG	random	20	0.650	0.604	0.169	0.233	0.285	0.260	0.336	0.985
MUTAG	GINE	GNNExpl.	random	20	0.550	0.823	0.452	0.372	−0.476	−0.006	−0.015	0.816
MUTAG	GINE	GuidedBP	random	20	0.550	0.823	0.258	0.268	−0.505	−0.007	−0.014	0.837
MUTAG	GINE	Input×Grad	random	20	0.550	0.823	0.054	0.181	−0.553	−0.008	−0.013	0.860
MUTAG	GINE	IG	random	20	0.550	0.823	0.048	0.180	−0.561	−0.009	−0.013	0.861
MUTAG	GINE	PGExpl.	random	20	0.550	0.823	**1.000**	1.000	−0.409	−0.003	−0.014	0.390
MUTAG	GINE	Saliency	random	20	0.550	0.823	0.020	0.175	−0.521	−0.008	−0.013	0.796
MUTAG	MPNN	IG	random	20	0.750	0.677	0.164	0.206	0.034	0.193	0.252	0.794
MUTAG	AttentiveFP	IG	scaffold	20	0.800	0.912	0.033	0.133	0.388	0.042	0.203	0.937
MUTAG	GAT	IG	scaffold	20	0.750	0.868	0.420	0.254	0.280	0.260	0.265	0.947
MUTAG	GCN	IG	scaffold	20	0.350	0.791	**0.950**	0.855	0.242	0.000	−0.002	0.628
MUTAG	GINE	GNNExpl.	scaffold	20	0.750	0.912	0.671	0.592	0.607	0.079	0.303	0.882
MUTAG	GINE	GuidedBP	scaffold	20	0.750	0.912	0.016	0.131	0.701	0.204	0.240	0.891
MUTAG	GINE	Input×Grad	scaffold	20	0.750	0.912	0.049	0.136	0.640	0.165	0.294	0.901
MUTAG	GINE	IG	scaffold	20	0.750	0.912	0.537	0.336	0.419	0.084	0.361	0.863
MUTAG	GINE	PGExpl.	scaffold	20	0.750	0.912	0.108	0.186	0.407	0.054	0.368	0.829
MUTAG	GINE	Saliency	scaffold	20	0.750	0.912	0.009	0.130	0.611	0.158	0.335	0.871
MUTAG	MPNN	IG	scaffold	20	0.750	0.879	0.672	0.473	**0.734**	0.229	0.258	0.909
SynthMotifs	AttentiveFP	IG	random	20	1.000	1.000	0.716	0.550	−0.162	0.335	0.296	0.655
SynthMotifs	GAT	IG	random	20	1.000	1.000	0.874	0.672	0.362	0.243	0.215	0.881
SynthMotifs	GCN	IG	random	20	0.600	1.000	0.480	0.339	0.376	0.119	0.142	0.693
SynthMotifs	GINE	GNNExpl.	random	20	1.000	1.000	0.474	0.250	0.022	0.171	0.222	0.018
SynthMotifs	GINE	GuidedBP	random	20	1.000	1.000	**1.000**	1.000	**0.640**	0.212	0.070	0.738
SynthMotifs	GINE	Input×Grad	random	20	1.000	1.000	0.992	0.968	0.568	0.212	0.084	0.633
SynthMotifs	GINE	IG	random	20	1.000	1.000	0.998	0.993	0.614	0.216	0.092	0.476
SynthMotifs	GINE	PGExpl.	random	20	1.000	1.000	0.490	0.167	−0.106	0.048	0.160	−0.014
SynthMotifs	GINE	Saliency	random	20	1.000	1.000	0.999	0.997	0.584	0.234	0.068	0.485
SynthMotifs	MPNN	IG	random	20	1.000	1.000	0.839	0.668	0.382	0.410	0.459	0.473
SynthMotifs	AttentiveFP	IG	scaffold	20	0.850	0.970	0.886	0.624	−0.121	0.276	0.247	0.805
SynthMotifs	GAT	IG	scaffold	20	0.950	0.990	0.592	0.433	0.155	0.071	0.066	0.337
SynthMotifs	GCN	IG	scaffold	20	0.700	1.000	**0.987**	0.932	0.325	0.269	0.164	0.787
SynthMotifs	GINE	GNNExpl.	scaffold	20	0.950	1.000	0.668	0.360	0.063	0.201	0.168	−0.008
SynthMotifs	GINE	GuidedBP	scaffold	20	0.950	1.000	0.928	0.728	0.194	0.209	0.095	0.833
SynthMotifs	GINE	Input×Grad	scaffold	20	0.950	1.000	0.968	0.881	0.207	0.242	0.109	0.716
SynthMotifs	GINE	IG	scaffold	20	0.950	1.000	0.901	0.655	0.292	0.281	0.060	0.657
SynthMotifs	GINE	PGExpl.	scaffold	20	0.950	1.000	0.565	0.329	0.116	0.184	0.164	0.268
SynthMotifs	GINE	Saliency	scaffold	20	0.950	1.000	0.975	0.915	0.194	0.256	0.099	0.610
SynthMotifs	MPNN	IG	scaffold	20	0.650	0.980	0.712	0.391	**0.525**	0.423	0.209	0.716
SynthMotifsXL [c]	GINE	GNNExpl.	random	120	1.000	1.000	0.501	0.259	0.033	0.364	0.472	—
SynthMotifsXL [c]	GINE	GuidedBP	random	120	1.000	1.000	**1.000**	0.998	**0.529**	0.495	0.006	—
SynthMotifsXL [c]	GINE	Input×Grad	random	120	1.000	1.000	0.989	0.958	0.393	0.493	0.089	—
SynthMotifsXL [c]	GINE	IG	random	120	1.000	1.000	0.987	0.945	0.403	0.493	0.021	—
SynthMotifsXL [c]	GINE	PGExpl.	random	120	1.000	1.000	0.269	0.165	−0.227	0.050	0.487	—
SynthMotifsXL [c]	GINE	Saliency	random	120	1.000	1.000	0.990	0.962	0.424	0.492	0.127	—

Data and code availability

All code, configurations, trained checkpoints, per-combination results, per-molecule audit records and the run manifest (seed, library versions, hardware, and the per-combination outcome ledger) are available at https://github.com/Kar488/molsanity, released under the MIT licence. A versioned archive of the release,

Table 7: Molecular classification cells (no node-level ground truth available), 1 of 2. Every committed classification cell on a real molecular dataset. The ground-truth localisation column does not exist for these datasets, since no per-atom labels are published, so only model-side reliability can be measured, which is the gap the Tier-1 cells are needed to close. *charact.* is the GraphFramEx characterisation score and *unfaith.* the PyG/DIG unfaithfulness metric, computed on the same molecules; both are blank for carried rows (^{c}), whose per-molecule records this run did not regenerate. Bold marks the highest occlusion ρ per dataset (emphasis only).

dataset	backbone	attributor	split	n	acc	AUC	ECE	motif top-1	occ. ρ	occ. top-1	Fid+	Fid−	stab.	charact.	unfaith.
BA-2Motifs	GINE	IG	random	100	0.820	1.000	0.076	0.209	−0.393	0.090	0.037	0.001	0.745	0.132	—
BA-2Motifs	GINE	IG	scaffold	100	0.990	0.990	0.253	0.119	**0.636**	0.360	0.022	0.029	0.740	0.044	—
BACE	GCN	IG	random	100	0.640	0.839	0.082	0.792	0.034	0.530	0.181	0.218	0.738	0.209	0.268
BACE	GCN	IG	scaffold	100	0.850	0.507	0.069	0.800	−0.668	0.140	−0.125	−0.125	0.898	0.067	0.276
BACE ^{c}	GINE	GNNExpl.	scaffold	30	0.433	0.434	0.189	0.835	−0.128	0.167	0.064	0.214	—	—	—
BACE ^{c}	GINE	Input×Grad	scaffold	30	0.433	0.434	0.189	0.841	−0.159	0.167	0.079	0.171	—	—	—
BACE	GINE	IG	random	100	0.730	0.846	0.055	0.844	0.307	0.640	0.347	0.278	0.844	0.366	0.219
BACE	GINE	IG	scaffold	100	0.340	0.651	0.179	0.843	**0.389**	0.810	0.104	0.092	0.571	0.184	0.066
BACE ^{c}	GINE	Saliency	scaffold	30	0.433	0.434	0.189	0.873	−0.174	0.167	0.097	0.173	—	—	—
BBBP	AttentiveFP	IG	random	100	0.770	0.855	0.050	0.750	**0.339**	0.470	0.164	0.167	0.844	0.234	0.203
BBBP	AttentiveFP	IG	scaffold	100	0.950	0.961	0.033	0.841	0.284	0.250	0.036	0.015	0.766	0.082	0.074
BBBP	GAT	IG	random	100	0.780	0.885	0.051	0.770	0.043	0.510	0.328	0.328	0.944	0.105	0.243
BBBP	GAT	IG	scaffold	100	0.980	0.805	0.010	0.831	−0.712	0.000	−0.021	−0.050	0.953	0.021	0.055
BBBP	GCN	IG	random	100	0.790	0.873	0.074	0.768	0.023	0.510	0.311	0.318	0.874	0.122	0.349
BBBP	GCN	IG	scaffold	100	0.980	0.966	0.013	0.814	−0.812	0.030	−0.010	−0.145	0.884	0.065	0.271
BBBP	GINE	GNNExpl.	random	100	0.820	0.886	0.053	0.779	−0.061	0.440	0.287	0.333	0.951	0.097	0.315
BBBP	GINE	GNNExpl.	scaffold	100	0.950	0.932	0.029	0.848	−0.368	0.060	0.018	−0.064	0.791	0.044	0.073
BBBP ^{c}	GINE	GuidedBP	scaffold	30	0.967	0.805	0.059	0.876	−0.530	0.000	−0.016	−0.037	—	—	—
BBBP ^{c}	GINE	Input×Grad	scaffold	30	0.967	0.805	0.059	0.839	−0.620	0.000	−0.016	−0.047	—	—	—
BBBP	GINE	IG	random	100	0.820	0.886	0.053	0.724	0.016	0.470	0.332	0.334	0.827	0.098	0.394
BBBP	GINE	IG	scaffold	100	0.950	0.932	0.029	0.841	−0.741	0.040	−0.031	−0.057	0.857	0.028	0.090
BBBP	GINE	PGExpl.	random	100	0.820	0.886	0.053	0.927	−0.050	0.310	0.086	0.332	0.562	0.075	—
BBBP	GINE	PGExpl.	scaffold	100	0.950	0.932	0.029	0.954	−0.538	0.060	−0.016	−0.060	0.488	0.031	—
BBBP ^{c}	GINE	Saliency	scaffold	30	0.967	0.805	0.059	0.826	−0.627	0.000	−0.029	−0.033	—	—	—
BBBP	MPNN	IG	random	100	0.840	0.914	0.053	0.792	0.154	0.420	0.210	0.267	0.765	0.121	0.104
BBBP	MPNN	IG	scaffold	100	0.950	0.922	0.033	0.824	−0.431	0.040	0.052	−0.043	0.832	0.117	0.096
ClinTox	GINE	GNNExpl.	random	100	0.710	0.910	0.157	0.504	0.059	0.430	0.258	0.230	0.984	0.108	0.323
ClinTox	GINE	GNNExpl.	scaffold	100	0.800	0.845	0.062	0.728	−0.242	0.390	0.139	0.135	0.983	0.128	0.289
ClinTox ^{c}	GINE	Input×Grad	scaffold	30	0.567	0.820	0.207	0.793	0.139	0.567	0.356	0.356	—	—	—
ClinTox	GINE	IG	random	100	0.710	0.910	0.157	0.754	−0.223	0.410	0.230	0.230	0.983	0.108	0.119
ClinTox	GINE	IG	scaffold	100	0.800	0.845	0.062	0.745	−0.329	0.380	0.135	0.135	0.991	0.128	0.254
ClinTox ^{c}	GINE	Saliency	scaffold	30	0.567	0.820	0.207	0.790	**0.145**	0.567	0.356	0.356	—	—	—

Table 8: Molecular classification cells (no node-level ground truth available), 2 of 2. Every committed classification cell on a real molecular dataset. The ground-truth localisation column does not exist for these datasets, since no per-atom labels are published, so only model-side reliability can be measured, which is the gap the Tier-1 cells are needed to close. *charact.* is the GraphFramEx characterisation score and *unfaith.* the PyG/DIG unfaithfulness metric, computed on the same molecules; both are blank for carried rows (^{c}), whose per-molecule records this run did not regenerate. Bold marks the highest occlusion ρ per dataset (emphasis only).

dataset	backbone	attributor	split	n	acc	AUC	ECE	motif top-1	occ. ρ	occ. top-1	Fid+	Fid–	stab.	charact.	unfaith.
DILI ^{c}	GINE	GNNExpl.	scaffold	30	0.800	0.824	0.141	0.826	−0.044	0.433	0.080	0.154	—	—	—
DILI ^{c}	GINE	Input×Grad	scaffold	30	0.800	0.824	0.141	0.858	−0.126	0.433	0.075	0.122	—	—	—
DILI	GINE	IG	random	48	0.688	0.719	0.135	0.760	0.298	0.500	0.354	0.360	0.780	0.383	0.393
DILI	GINE	IG	scaffold	48	0.688	0.824	0.120	0.852	**0.346**	0.521	0.194	0.229	0.941	0.276	0.428
DILI ^{c}	GINE	Saliency	scaffold	30	0.800	0.824	0.141	0.851	−0.128	0.433	0.056	0.134	—	—	—
SIDER	GCN	IG	random	100	0.640	0.677	0.050	0.722	0.479	0.720	0.141	0.249	0.782	0.220	0.048
SIDER	GCN	IG	scaffold	100	0.650	0.663	0.054	0.746	0.067	0.620	0.060	0.118	0.831	0.144	0.062
SIDER ^{c}	GINE	GNNExpl.	scaffold	30	0.633	0.661	0.110	0.816	0.490	0.833	0.038	0.196	—	—	—
SIDER ^{c}	GINE	Input×Grad	scaffold	30	0.633	0.661	0.110	0.878	0.576	0.867	0.088	0.196	—	—	—
SIDER	GINE	IG	random	100	0.710	0.682	0.049	0.761	0.516	0.660	0.157	0.315	0.785	0.221	0.452
SIDER	GINE	IG	scaffold	100	0.570	0.650	0.072	0.836	**0.816**	0.860	0.065	0.082	0.661	0.139	0.490
SIDER ^{c}	GINE	Saliency	scaffold	30	0.633	0.661	0.110	0.887	0.571	0.867	0.093	0.201	—	—	—
Tox21 ^{c}	GINE	GNNExpl.	scaffold	30	1.000	0.827	0.054	0.719	**0.010**	0.367	−0.066	−0.061	—	—	—
Tox21 ^{c}	GINE	Input×Grad	scaffold	30	1.000	0.827	0.054	0.704	−0.409	0.167	−0.075	−0.060	—	—	—
Tox21	GINE	IG	random	100	0.970	0.849	0.099	0.752	−0.739	0.150	−0.058	−0.080	0.639	0.017	0.205
Tox21	GINE	IG	scaffold	100	0.960	0.822	0.028	0.739	−0.367	0.250	−0.011	−0.009	0.381	0.031	0.188
Tox21 ^{c}	GINE	Saliency	scaffold	30	1.000	0.827	0.054	0.697	−0.391	0.167	−0.073	−0.063	—	—	—
hERG ^{c}	GCN	IG	scaffold	30	0.900	0.802	0.098	0.665	**0.711**	0.867	0.589	0.590	—	—	—
hERG ^{c}	GINE	GNNExpl.	scaffold	30	0.733	0.784	0.070	0.725	0.066	0.633	0.094	0.408	—	—	—
hERG ^{c}	GINE	Input×Grad	scaffold	30	0.733	0.784	0.070	0.633	0.342	0.600	0.410	0.408	—	—	—
hERG	GINE	IG	random	66	0.803	0.824	0.093	0.776	0.661	0.833	0.560	0.560	0.992	0.279	0.659
hERG	GINE	IG	scaffold	66	0.727	0.805	0.070	0.689	0.245	0.606	0.337	0.337	0.924	0.283	0.527
hERG ^{c}	GINE	Saliency	scaffold	30	0.733	0.784	0.070	0.626	0.343	0.533	0.409	0.408	—	—	—

including the trained weights, is deposited with a DOI [*DOI to be inserted on deposition*]; the repository carries `CITATION.cff` and `.zenodo.json` so the archive is generated from a tagged release.

Every figure and table in this paper is generated from those artifacts by `paper/figs/make_figures.py` and `paper/figs/make_tables.py`, which read the released results directly; no value in this manuscript is transcribed by hand. Re-running them after a new sweep updates every number, figure and table in place. Dataset licences and provenance are recorded in the repository's dataset manifest; the credential-gated resources referenced in Section Experimental setup are out of scope and were never accessed.

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